Data Science Internship Projects at CodeAlpha

A Summer Training Project Report Submitted in Partial Fulfillment of the Requirements for the Award of Degree

Of

Bachelor of Technology

In

Computer Science and Engineering

By

Rakesh Kumar (11232952)

Under the supervision of

Dr. Misha Mittal

Assistant Professor

M. M. Engineering College, Mullana, Ambala

Maharishi Markandeshwar (Deemed to be University), Mullana, Ambala, Haryana, India

November 2025

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Department of Computer Science & Engineering

M. M. Engineering College, Mullana, Ambala, Haryana, India

Candidate's Declaration

I hereby certify that the work which is being presented in this project entitled **"Data Science Internship Projects at CodeAlpha"**, in fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** of Maharishi Markandeshwar (Deemed to be University), Mullana, Ambala, Haryana, India, is an authentic record of my own work carried out during a period from **May 2025 to June 2025** under the supervision of **Dr. Misha Mittal (Assistant Professor)**. The matter presented in this project report has not been submitted by me for the award of any other degree of this or any other University/Institute.

Rakesh Kumar (11232952)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Supervisor's Name: - Dr. Misha Mittal

Designation: - Assistant Professor, Department of CSE

Professor Signature: - …………..………..

Date: - ……………………………………….

Department of Computer Science & Engineering

M. M. Engineering College, Mullana, Ambala, Haryana, India

Acknowledgement

I would like to express my heartfelt gratitude to my guide **Dr. Misha Mittal**, Department of Computer Science and Engineering, M.M. Engineering College, Mullana, for her valuable guidance, encouragement, and constant support throughout the internship period. I am also thankful to **CodeAlpha**, for providing me with this wonderful opportunity to work as a **Data Science Intern** from **1st June 2025 to 30th June 2025**. This internship gave me practical exposure to various Machine Learning and Data Science technologies.

I also express my sincere thanks to **Dr. Sandip Kumar Goel**, Professor and Head of the CSE Department, for his continuous motivation and providing a research-oriented environment in the department.

Lastly, I am deeply grateful to my **parents and friends** for their constant moral support and encouragement which motivated me to complete this project successfully.

Rakesh Kumar (11232952)

Abstract

This report presents the work completed during my Data Science Virtual Internship at CodeAlpha. The internship aimed to develop a better understanding of real-world applications of Data Science and Machine Learning using Python.
During the internship period, I successfully completed three projects:

1. Car Price Prediction using Machine Learning
2. Iris Flower Classification
3. Unemployment Analysis using Python

Each project helped me to apply theoretical knowledge in a practical way — performing data preprocessing, visualization, model building, and evaluation. These projects enhanced my ability to handle datasets, clean data, train models, and interpret predictions. The report covers the methodology, algorithms used, implementation, and future enhancements of the projects. This internship experience significantly strengthened my technical foundation in Python, Data Analysis, and Machine Learning.

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CHAPTER -1

INTRODUCTION ABOUT COMPANY (CODE ALPHA)

1.1 Company Profile

CodeAlpha is a reputed virtual internship and training platform that provides students and professionals with opportunities to gain practical skills in fields like Web Development, Artificial Intelligence, Machine Learning, Data Science, and Cyber Security.

1.2 Objective of CodeAlpha

The main objective of CodeAlpha is to bridge the gap between theoretical education and practical industrial experience by offering real-world projects.

1.3 Technologies Used in CodeAlpha Internship

The technologies used during the internship include:

- Python Programming Language
- Machine Learning Algorithms
- Data Visualization Tools (Matplotlib, Seaborn)
- Jupyter Notebook Environment

1.4 Internship Details

- Internship Period: 1st June 2025 – 30th June 2025
- Internship Type: Virtual (Remote)
- Domain: Data Science
- Organization: CodeAlpha

1.5 Company Website: www.codealpha.tech

CHAPTER 2:

OVERVIEW OF INTERNSHIP TRAINING

During my one-month internship at CodeAlpha, I received comprehensive training in the field of Data Science, gaining hands-on experience with tools and technologies commonly used in real-world data analysis and machine learning. The program emphasized the practical understanding of data preprocessing, model building, and performance**.**

2.1 Technologies and Tools Learned

Throughout the training, I worked with the following technologies:

- Python Programming Language
- NumPy and Pandas for data manipulation and analysis
- Matplotlib and Seaborn for data visualization
- Scikit-learn for implementing machine learning algorithms
- Jupyter Notebook as the main environment for coding and experimentation

2.2 Project Work

The projects undertaken during the internship were selected to cover essential aspects of Data Science and Machine Learning:

1. Car Price Prediction – A regression-based project to predict vehicle prices.
2. Iris Flower Classification – A classical classification problem using supervised learning.
3. Unemployment Analysis – A real-world dataset analysis with visual insights into employment trends.

2.3 System and Platform Requirements

- Operating System: Windows 10 or later
- Software: Python 3.11, Jupyter Notebook
- Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- Hardware: Minimum 8 GB RAM and 250 GB Storage

Outcome

This internship phase significantly enhanced my technical proficiency and analytical thinking, enabling me to apply Data Science concepts to real-world problems effectively. I developed a deeper understanding of data handling, model development, and predictive analytics, which strengthened my foundation in the field.

CHAPTER 3:

INTRODUCTION & OBJECTIVES OF PROJECT WORK

The primary objective of my internship at CodeAlpha was to gain practical exposure to the complete workflow of Data Science, from raw data collection and preprocessing to model development, evaluation, and visualization. The internship provided hands-on experience in transforming complex datasets into valuable insights using analytical and machine learning techniques.

Through this training, I understood how organizations rely on data-driven decision-making in today's technology-driven world. The internship enhanced my knowledge of data preprocessing, feature engineering, model selection, and performance evaluation, and strengthened my programming and analytical skills in Python.

By the end of the internship, I had developed the ability to handle real-world data challenges, clean and prepare messy datasets, visualize data trends, and train machine learning models capable of producing accurate predictions and classifications.

3.1 Objectives

The major objectives of this internship were:

- To understand the process of designing, implementing, and optimizing Machine Learning models.
- To gain practical experience in data preprocessing, including handling missing data, encoding categorical variables, and normalizing datasets.
- To learn and apply techniques for data visualization using graphs, charts, and heatmaps to derive meaningful insights.
- To develop predictive and classification models that can address real-world business and analytical problems.
- To analyze model performance using statistical metrics and interpret results effectively.
- To strengthen my proficiency in Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

This internship acted as a bridge between the theoretical foundations learned in academics and their industrial applications, helping me apply data science methodologies to solve real-world problems efficiently.

3.2 Projects Undertaken

During the internship, I completed three key projects under the guidance of my mentor. Each project was designed to explore a different phase of the **Data Science lifecycle**, including data analysis, regression modelling, and classification.

1. **Car Price Prediction using Machine Learning**
    - Objective: To develop a regression model for predicting car prices based on attributes such as year of manufacture, fuel type, seller type, and current price.
    - Outcome: Successfully implemented a model capable of accurately estimating vehicle resale values.
2. **Iris Flower Classification**
    - Objective: To classify iris flowers into three categories—Setosa, Versicolor, and Virginica—using supervised learning algorithms such as Logistic Regression and Decision Tree.
    - Outcome: Enhanced understanding of classification algorithms, model evaluation, and accuracy comparison.
3. **Unemployment Analysis using Python**
    - Objective: To analyze unemployment trends across various Indian states and visualize regional variations using data visualization tools.
    - Outcome: Gained practical experience in data cleaning, visualization, and statistical interpretation of socio-economic datasets.
    - This internship experience at CodeAlpha not only improved my technical competence but also helped me develop a data-driven mindset, preparing me for future roles in the field of Data Science and Artificial Intelligence.

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PROJECT-1

CAR PRICE PREDICTION USING MACHINE LEARNING

4.1 Introduction

The objective of this project was to predict the selling price of used cars based on various attributes such as the manufacturing year, present price, fuel type, seller type, and ownership details. This project illustrates the application of regression models in solving real-world problems where the target variable is continuous in nature. By analyzing historical car data, the model helps estimate the resale value of a car with improved accuracy.

4.2 Methodology

The project followed a systematic Data Science workflow consisting of the following steps:

1. **DataCollection:**
   The dataset was obtained from Kaggle (Car Data), containing relevant features such as car model, year, fuel type, and selling price.

2. **Data Preprocessing:**
   - Removed missing or duplicate entries.
   - Encoded categorical features (like fuel type and seller type) into numerical values.
   - Performed feature selection to identify the most influential factors affecting price.

3. **Model Development:**
   - Implemented and trained machine learning models such as Linear Regression and Random Forest Regressor to predict car prices.

4. **Model Evaluation:**
   - Evaluated model performance using metrics such as R^2 Score and Mean Absolute Error (MAE).
   - Compared results from different algorithms to select the best-performing model.

4.3 Algorithm Used

The Linear Regression Algorithm was used to find relationships between dependent (Price) and independent variables.

4.4 Tools Used

- Programming Language: Python
- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
- Development Environment: Jupyter Notebook

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4.5 Results

After model training and evaluation, the developed regression model achieved **high prediction accuracy**, effectively estimating the selling price of cars. The results demonstrated that car age and current market price are the most significant predictors of resale value.

Sample Results:

Car Model	**Year**	**Present Price**	**Predicted Selling Price**
Honda city	2018	8.5	6,9
Toyota Innova	2015	12,0	8.2

4.6 Discussion

The model performed well on unseen data and provided realistic predictions. Visualizations showed strong correlation between "Present Price" and "Year" features with selling price.

4.6 Future Scope

In the future, this project can be enhanced and deployed as a web application using frameworks such as Flask or Streamlit, allowing users to input car details and instantly receive estimated resale prices.
Further improvements can include integration of larger datasets, feature scaling, and use of advanced algorithms such as Boost or Neural Networks for improved accuracy.

PROJECT -2

IRIS FLOWER CLASSIFICATION

5.1 Introduction

The Iris Flower Classification project is based on one of the most well-known datasets in the field of Machine Learning — the Iris dataset. The primary goal of this project was to classify iris flowers into three species: *Setosa*, *Versicolor*, and *Virginica*, using their sepal length, sepal .This project demonstrates the application of supervised classification algorithms in recognizing patterns and making accurate predictions from numeric data..

5.2 Methodology

1. Data Import: Used the inbuilt Iris dataset from Scikit-learn.
2. Data Visualization: Plotted Petal Length, Petal Width, Sepal Length, and Sepal Width to understand distribution.
3. Model Training: Used Logistic Regression and Decision Tree Classifier algorithms.
4. Evaluation: Accuracy score and confusion matrix were used to evaluate performance.

5.3 Algorithm Used

- **Logistic Regression:** Applied to classify flowers based on continuous numerical input features. It provided a strong baseline for linear separability.
- **Decision Tree Classifier:** Used to capture non-linear relationships and visualize the decision-making process through a tree structure.
- .

5.4 Tools Used

- **Programming Language:** Python
- **Libraries:** Scikit-learn, Matplotlib, Seaborn, Pandas, NumPy
- **Platform:** Jupyter Notebook

5.5 Results

The developed models achieved over 96% classification accuracy, effectively distinguishing between the three flower species. The confusion matrix confirmed the model's strong generalization capability on unseen data

Confusion Matrix Output:

Predicted / Actual	Setose	Versicolor	Virginica
Setose	49	1	0
Versicolor	0	46	4
Virginica	0	2	48

5.6 Discussion

The classification model performed exceptionally well, with clear separation boundaries observed between species. Visualizations indicated that *Petal Length* and *Petal Width* were the most significant features for distinguishing between the three classes. The Decision Tree model provided an interpretable representation of how feature thresholds influence classification outcomes.5.7 Future Scope

Future enhancements can include:

- Implementing Deep Learning models such as Neural Networks for improved feature learning.
- Integrating the trained model into a mobile or web application for educational and botanical research use.
- Extending the approach to classify other flower species using transfer learning techniques.

PROJECT -3

UNEMPLOYMENT ANALYSIS IN INDIA

6.1 Introduction

This project aimed to analyze the unemployment rate in India during and after the COVID-19 pandemic, focusing on identifying trends, regional disparities, and gender-based variations. The

analysis provides valuable insights into how the pandemic affected the employment landscape and economic stability across the country.

6.2 Methodology

1. Dataset:
   Collected from Kaggle's "Unemployment in India" dataset, which contains region-wise and gender-wise unemployment statistics.

2. Data Cleaning:
   - Removed missing and duplicate entries.
   - Standardized region names for consistency.
   - Converted date fields into proper formats for time-based analysis.

3. Data Visualization:
   - Created bar graphs, line charts, and heatmaps using Matplotlib and Seaborn to explore unemployment trends over time.

4. Analysis:
   - Compared unemployment rates by state, region type (urban/rural), and gender.
   - Identified periods of high unemployment corresponding to COVID-19 lockdown phases.

6.3 Tools Used

- Programming Language: Python
- Libraries: Pandas, Matplotlib, Seaborn, NumPy
- Development Environment: Jupyter Notebook

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6.4 Results

- The unemployment rate peaked during 2020 due to the lockdown period.
- States like Bihar and Jharkhand showed higher unemployment rates compared to others.
- Female unemployment was slightly higher in rural regions.

Sample Visualization Insight:

Maharashtra – 8.2%

Bihar – 12.5%

Karnataka – 5.6%

Tamil Nadu – 6.3%

Jharkhand-11.4%

6.5 Discussion

Through visual analysis, the data highlighted the correlation between region type and unemployment rate.
This analysis could help policymakers identify target areas for job creation programs.

6.6 Future Scope

- Integration of real-time unemployment data APIs for dynamic trend tracking.
- Development of interactive dashboards using Power BI or Tableau for more engaging visual exploration.
- Expanding the study to include education level and industry sector analysis for deeper socioeconomic insights.

CHAPTER -6

FUTURE SCOPE OF INTERNSHIP & LEARNING OUTCOME

7.1 Future Scope

The internship projects have significant potential for future enhancement and expansion. The following improvements can be implemented:

- Web Deployment: Deploy the developed machine learning models on interactive platforms using Flask or Streamlet, enabling real-time user predictions.
- Dataset Expansion: Integrate multiple and larger datasets to enhance model generalization and prediction accuracy.
- Visualization Dashboards: Build dynamic dashboards using tools like Power BI or Tableau for live data monitoring and trend analysis.

7.2 Learning Outcome

The internship helped me in:

- Gaining hands-on experience with Python and ML libraries.
- Understanding the end-to-end Data Science workflow.
- Improving logical thinking and debugging skills.
- Learning real-world problem-solving through data-driven insights.

Overall, this internship has been an enriching experience that has prepared me for future research and industrial projects in Machine Learning.

This internship has been a transformative learning experience, allowing me to apply theoretical knowledge to real-world problems. I gained hands-on experience with Python programming, data analysis, and machine learning libraries such as Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn. I also learned how to manage data preprocessing, visualization, model training, and evaluation in a structured workflow.

Beyond technical skills, this internship strengthened my logical reasoning, debugging, and analytical thinking abilities. Working on multiple datasets improved my understanding of model accuracy, overfitting, and evaluation metrics. It also helped me appreciate the importance of clean data and the challenges faced while working with real-world information.

References

- Kaggle Datasets: Car Price, Iris, Unemployment in India.
- Python Documentation – www.python.org
- Scikit-learn Documentation – https://scikit-learn.org
- Matplotlib Documentation – https://matplotlib.org
- CodeAlpha Internship Resources – www.codealpha.tech

Appendix A: Source Code

Project-3: Car Price Prediction

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from xgboost import XGBRegressor

from sklearn import metrics

import sklearn.datasets

import kagglehub

path = kagglehub.dataset_download("vijayaadithyanvg/car-price-predictionused-cars")

print("Path to dataset files:", path)

import os

os.listdir('/root/.cache/kagglehub/datasets/vijayaadithyanvg/car-price-predictionused-cars/versions/1')

df = pd.read_csv('/root/.cache/kagglehub/datasets/vijayaadithyanvg/

car-price-predictionused-  cars/versions/1/car data.csv')

df.head()

df.shape

df.isnull()

correlation = df.corr(numeric_only=True)
```

```
plt.figure(figsize=(10,10))
sns.heatmap(correlation,cbar=True,square=True,fmt='.1f',annot=True,annot_kws={'size':8},cmap='Blues')
df = pd.get_dummies(df,columns=['Fuel_Type','Selling_type','Transmission'],drop_first=True)
df = df.drop(['Car_Name'],axis=1)
X = df.drop(['Selling_Price'],axis=1)
Y = df['Selling_Price']
X_train,X_test,Y_train,Y_test = train_test_split(X,Y,test_size=0.2,random_state=42)
print(X.shape,X_train.shape,X_test.shape)
model = XGBRegressor()
model.fit(X_train,Y_train)
training_data_prediction = model.predict(X_train)
score_1 = metrics.r2_score(Y_train,training_data_prediction)
score_2 = metrics.mean_absolute_error(Y_train,training_data_prediction)
print('R Sqaured Error:',score_1)
print('Mean Absolute Error:',score_2)
plt.scatter(Y_train,training_data_prediction)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual Price vs Predicted Price")
plt.show()
test_data_prediction = model.predict(X_test)
score_1 = metrics.r2_score(Y_test,test_data_prediction)
score_2 = metrics.mean_absolute_error(Y_test,test_data_prediction)
print('R Sqaured Error:',score_1)
print('Mean Absolute Error:',score_2)
```

Project 2: Iris Flower Classification

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt, plotly.express as px
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score,confusion_matrix,classification_report

import pickle

df=pd.read_csv("Iris.csv")

X=df.drop(["Id","Species"],axis=1)

y=LabelEncoder().fit_transform(df["Species"])

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)

model=RandomForestClassifier(random_state=42)

model.fit(X_train,y_train)

y_pred=model.predict(X_test)

print("Accuracy:",accuracy_score(y_test,y_pred))

print("Confusion Matrix:\\n",confusion_matrix(y_test,y_pred))

print("Report:\\n",classification_report(y_test,y_pred))

fig=px.scatter(df,x="SepalWidthCm",y="SepalLengthCm",color="Species",title="Sepal Width vs
Sepal Length"); fig.show()

pickle.dump(model,open("Iris_Model.pkl","wb"))
```

Project 3: Unemployment Analysis in India

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns, plotly.express as px

df=pd.read_csv("Unemployment_Rate_upto_11_2020.csv")

df[' Date']=pd.to_datetime(df[' Date'],errors='coerce')

df.dropna(inplace=True)

plt.figure(figsize=(12,6))

sns.lineplot(x=' Date',y=' Estimated Unemployment Rate (%)',data=df,color='blue')

plt.title('Unemployment Rate Over Time'); plt.xlabel('Date'); plt.ylabel('Rate (%)');
plt.tight_layout(); plt.show()

if 'Region' in df.columns:

    plt.figure(figsize=(14,6))

    sns.barplot(x='Region',y=' Estimated Unemployment Rate (%)',data=df)

    plt.title('Average Unemployment Rate by Region'); plt.xticks(rotation=45); plt.show()
```

```
fig=px.scatter(df,x=' Date',y=' Estimated Unemployment Rate (%)',color='Region' if 'Region' in df.columns else None,
    title='Unemployment Rate Over Time (Interactive)'); fig.show()
print("Avg Rate:",round(df[' Estimated Unemployment Rate (%)'].mean(),2),"%")
print("High:",round(df[' Estimated Unemployment Rate (%)'].max(),2),"%")
print("Low:",round(df[' Estimated Unemployment Rate (%)'].min(),2),"%")
```

Appendix B: Abbreviations

- ML – Machine Learning
- AI – Artificial Intelligence
- CSV – Comma Separated Values
- DFD – Data Flow Diagram

INTERNSHIP CERTIFICATE

CODE ALPHA

YOUR DREAM OUR PASSION

Student ID:CA/JU1/2267

CERTIFICATE

OF COMPLETION

This Certificate Is Proudly Presented To

Rakesh Kumar

Was an active Participant at CodeAlpha Virtual Internship Program in Data Science Internship with dedication and hard work. We really appreciate your efforts taken and wish you all the best for the further.

1st June 2025 to 30th June 2025

CEO & Founder

2nd July 2025

Date of Issue

MSME

MINISTRY OF MICRO, SMALL & MEDIUM ENTERPRISES

LETTER OF RECOMMENDATION

OFFER LETTER

CODE
ALPHA
YOUR DREAM OUR PASSION

INTERNSHIP OFFER LETTER

17th May 2025 **STUDENT ID: CA/JU1/2267**

Name: Rakesh kumar

Welcome to CodeAlpha

Congratulations**!** We are delighted to make you an offer as **Data Science Internship** offer letter is to confirm your selection **Data Science Internship** at **CodeAlpha** and welcome you for the same effective from 1st June 2025 to 30th June 2025. Hope you will be doing well.

This Internship is observed by CodeAlpha as being a learning opportunity for you.

Your internship will embrace orientation and give emphasis on learning new skills with a deeper understanding of concepts through hands-on application of the knowledge you gained as an intern. You will acknowledge your obligation to perform all work allocated to you to the best of your ability within lawful and reasonable direction given to you.

We look forward to a worthwhile and fruitful association which will make you equipped for future projects.

Wishing you the most enjoyable and truly meaningful internship program experience.

Sincerely,

Swati Sri

Swati Srivastava
Founder & CEO

CODEALPHA CODE ALPHA SOFTWARE DEVELOPMENT

GOVERNMENT OF INDIA
MSME
MINISTRY OF MICRO, SMALL & MEDIUM ENTERPRISES

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